

explains. “These features will enable deployment of AI in a frugal and sustainable way in India.”

Why does that matter so much? Because as AI expands beyond research labs and top-tier data centers, it starts trickling into smaller businesses, municipalities, and edge devices. If the hardware is too power-hungry, which is usually the case with GPUs (as any gamer will agree) you can bet it won't scale seamlessly in a region where resources can be more constrained, especially outside major metros.

India's AI ecosystem: Rich in data and talent pool

It's not just about hardware. India is bursting at the seams with data – and data is, after all, the lifeblood of AI. According to Santhosh, “India generates nearly 20% of the world's data,” an astounding figure that underscores why so many tech giants are eager to plant deeper roots here. He also believes the vast local talent base is a key advantage, and Intel is keen on nurturing it through global system integrators and independent software vendors.

He name-drops an interesting collaboration with Infosys on an AI-first suite of solutions called Infosys Topaz, leveraging Intel Xeon processors and Gaudi 2. Another one with Bud Ecosystem for making generative AI inference possible on CPUs. “Bud has also trained and optimized foundational models on Gaudi, creating real-time private agents that can run on AI PCs.”

In addition, Santhosh also points out a couple of interesting local collaborations that highlight what Intel sees as a blueprint for the future. “There are many great AI examples in India, including Intel's collaboration with Haystack Analytics to create an edge compute solution for faster tuberculosis testing.” He also mentions a Generative AI model designed to converse in many Indic languages, done in partnership with Tech Mahindra.

“We're expanding AI education and digital readiness programs,” he adds. Indeed, Intel has taught 1.6 million

students under its AI for Youth initiative and offered AI awareness to over 4.5 million citizens via AI for All – two programs that, while massive in scope, might just be scratching the surface of India's potential.

As far as Santhosh or Intel India is concerned, these partnerships are more than just business deals – they're seeds for an ever-expanding AI ecosystem that's starting to flourish in India. Because the potential payoff is huge, as IDC's Asia/Pacific AI Maturity Study 2024 suggests India's AI spending could hit \$5 billion by 2027. That's why Intel's strategic approach is focused on putting the right hardware in place, developing local competencies, and relying on a supportive regulatory environment to accelerate economic and technological gains.

Make in India and Intel's evolving role

Now, with a national push for “Make in India,” the question is how Intel's role might morph, especially since it's had a local presence in the country for over three decades, particularly through design centre investments. In case you didn't know, a big part of Intel's India operation is about design and engineering – all through Intel's software and hardware products and services stack – supported by 12,000 employees and state-of-the-art facilities. “We've collaborated with policymakers, government bodies, and ecosystem partners to grow the electronics manufacturing sector,” Santhosh says. They're providing early access to products, reference designs, and even supporting local manufacturing of HPC servers.

But the path isn't smooth. He laments how India has historically been services-focused, whereas scaling manufacturing means developing strong product skills. Tying in the dynamic regulatory framework – often essential but occasionally adding complexity – Santhosh stresses the need for stable infrastructure like power and logistics. He believes India needs to

work in two key areas going forward in order to scale up production. “Two key factors are critical for manufacturing at scale. The first is demand – without a strong domestic market, local production struggles. The second is innovation. We need to strengthen design and innovation capabilities, going beyond cost optimization.”

Santhosh points out that cost disadvantages and the need for a robust component ecosystem also remain key stumbling blocks. “We're committed to upskilling the workforce, driving tech advancements, and ensuring India-specific designs can compete globally,” he insists, painting a picture where Intel is both a beneficiary and an enabler of India's aspirations to become a heavyweight in electronics manufacturing. “Intel remains committed,” he adds, referencing the company's willingness to work closely with government and industry to expedite advanced technology manufacturing.

Additionally, Santhosh sees India as a vital node in a more balanced global supply chain, especially relevant in a post-pandemic world wary of single-region dependencies. “The Make in India initiative is pivotal in fostering a diversified and resilient electronics manufacturing ecosystem,” he says, citing record-breaking \$115 billion electronics output in FY24, propelled by mobile phone production.

He calls out the revised IT hardware PLI scheme and new proposals for electronic components as strong incentives, aiming to capture \$500 billion in electronics production by 2030. It's a big vision, but not without hurdles, as mentioned above. If done right, though, “India could become a key player in the global supply chain, boosting resilience and self-reliance,” he emphasises. **d**

